

# Adharsh Shokkalingam

Phone: 825-440-2394 | Email: [Adharsh2022@icloud.com](mailto:Adharsh2022@icloud.com) | [LinkedIn](#) | [Github](#)

## Education

University of Alberta

Bachelor of Science in Computer Science, Minor Mathematics

Expected Graduation: May 2027

**Technical Skills:** Python, Java, C/C++, SPSS, RISC-V Assembly (RV32I), HTML, CSS

**Developer Tools:** Git, VS Code, FireBase, UML, Android Studio, SDL2, SQL, MongoDB, Figma

**Relevant Coursework:** Algorithms, Software Engineering, Computer Architecture, Operating Systems, Machine Learning, Databases.

## Work Experience

- **Sports Facility Attendant | Edmonton Scottish Soccer Dome** *June 2023 - Current*
  - Assist in the setup and breakdown of equipment and facilities for various sports activities, events, and programs. Enforce facility rules and regulation to ensure the safety of patrons.
  - Maintain cleanliness and organization of the facility, including locker rooms, Dome, and common areas, to uphold sanitation standards and enhance the overall atmosphere.
  - Collaborate closely with coworkers and supervisors to coordinate setups, resolve issues in real time, and ensure events and daily operations run smoothly and on schedule.

## Projects

- **3D-Engine CPU Rasterizer - C, SDL2** *June 2025 - August 2025*
  - Engineered a fully interactive **3D engine** in **C**, leveraging **SDL2** for graphics abstraction and input handling, with backface culling, near plane clipping, dynamic camera transformation, and basic lighting.
  - Designed multiple **real-time rendering modes** (**wireframe, vertex-highlighted, solid-fill, textured**) with keyboard-driven mode switching and **6** degrees of camera freedom.
  - Architected the project into **modular components** (**lighting, geometry, mesh, culling, camera, etc**), **enabling extensibility and low level control over the render pipeline.**
- **Sulfur - Events Lottery Android App - Java, Android Studio, FireBase** *Sept 2025- Dec 2025*
  - Developed a full stack **android** application with **6** group members using **Java, android studio, Firebase** for managing and **tracking** event/user data.
  - Implemented **QR Code** generation and scanning features including the storing and extraction of embedded **geolocation** data to display and update event/users location.
  - Collaborated in an Agile environment with **Git**, participated in code reviews, and improved/refined application features.
  - Co-designed the application's **UML (class, sequence/activity, and data model diagrams)** architecture to standardize module boundaries and ensure **maintainable code structure.**
  - Utilized **Firebase Cloud Firestore** for structured cloud data storage, **querying**, and **multi-user** data consistency.
  - Built the capacity based lottery workflow (**sample N entrants** after the registration window, notifications, re-sampling on declines) to ensure fair sign-ups.
- **RISC-V to WASM Converter - RISC-V (RV32I) (CMPUT 229 - Lab)** *Sept 2025- Dec 2025*
  - Implemented a **Binary** level translator converting **RISC-V** Machine instructions to **WASM** Byte code.
  - Designed and implemented support for **R-type, I-type, and Branch instructions**, including arithmetic, logic, shifts and **conditional control flow.**
  - Built a **signed LEB128 encoder** from scratch to extract immediate operands and convert to **WASM** byte code.
  - Mapped **RISC-V** registers to **WASM** locals using a **custom register translation table** and handled architectural differences such as the **zero register.**